Object Oriented Programming (CST-004) - Unit 1

Unit 1 - OOP Concepts, Classes & Pointers - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: the four pillars of OOP, how C++ classes and objects work, and pointers - the foundation for everything after.

Key ideas: Encapsulation = data and functions bundled in one class; **abstraction** = show what, hide how; **inheritance** = reuse by deriving classes; **polymorphism** = same name, many behaviours. A **class** is a blueprint, an **object** its instance. Access specifiers: **private** (default - only inside the class), **public** (anywhere), **protected** (class + derived classes). A **pointer** stores an address: * declares/dereferences, & takes an address. **new**/**delete** manage heap memory, the **this** pointer points to the calling object. A **reference** is an alias - must be initialised and cannot be reseated.

Exam pointers: "explain the four pillars with examples" is the classic theory question. Expect small code outputs using pointers (* and &) and a class with private data accessed through public member functions. Know the default specifier: class = private, struct = public.